

CoRoT-2b

R-0024

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_260526 · created 2026-07-08T10:35:36+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461186.871 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.223 +/- 0.065
Transit depth (Rp/Rs)^2	4.99 +/- 2.89 [%]
Orbital Inclination (inc)	85.39 +/- 2.09
Airmass coefficient 1 (a1)	1.9 +/- 0.83
Airmass coefficient 2 (a2)	-0.35 +/- 0.16
Scatter in the residuals of the lightcurve fit is	43.7 %
Best Comparison Star	None
Optimal Aperture	3.26
Optimal Annulus	4.34
Transit Duration (day)	0.088 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.223 ± 0.065 vs 0.1667 (deviation 0.77σ)
Duration fit vs published	0.088 d vs 0.095 d (deviation 0.41σ)
Mid-time O–C	7.9 min
Depth SNR	3.0
Residual scatter	43.7 %

Light curve

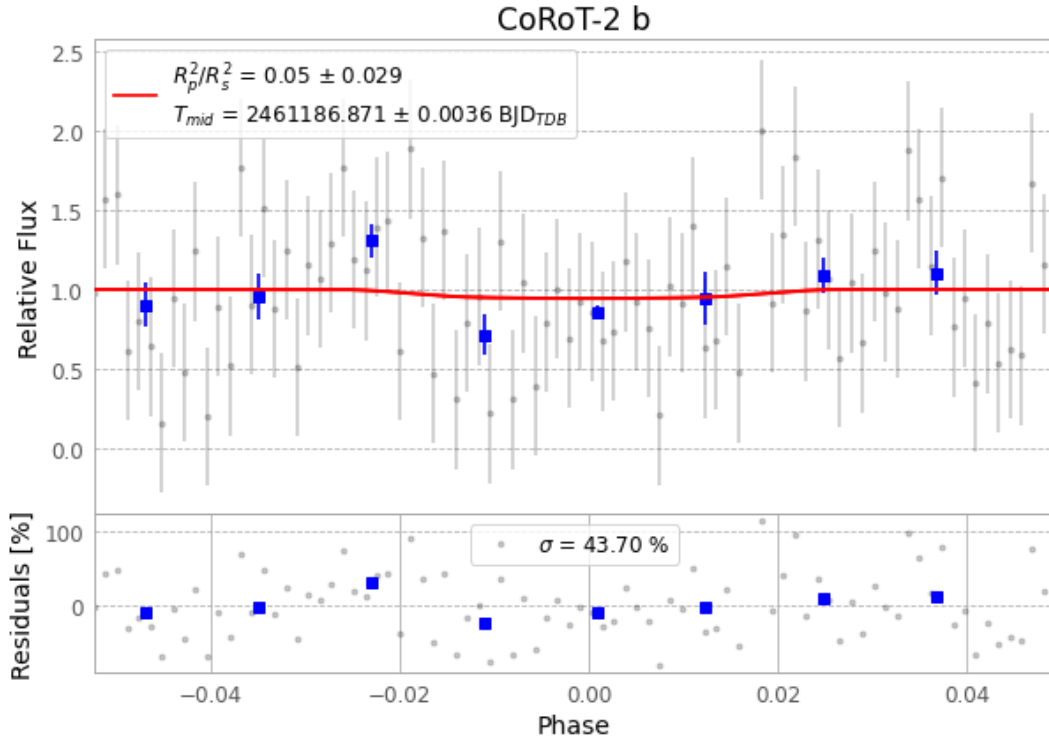


Figure 1 — FinalLightCurve_CoRoT-2 b_2026-05-26.png (SHA-256 3e5382ba7768e815...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [260, 191], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f7e7124a2d6054e6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

86 FITS frames (56 MB), CoRoT-2260526063615.FITS → CoRoT-2260526105115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-260526.log (af304a83e01e...), FinalLightCurve_CoRoT-2 b_2026-05-26.png (3e5382ba7768...), FinalLightCurve_CoRoT-2 b_2026-05-26.csv (c4ad983d0794...)

Compute resources

Wall time 147.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.